

BSc

MECHANICAL ENGINEERING

Mechanical Engineering at ACity

The creatively formed curriculum gives students a holistic overview of how mechanical systems work together to produce machines. They are given the fundamental principles, to start off on an experiential learning trajectory, providing students with hands-on development of sample systems in real-life design-and-build projects.

Our goal is to produce ethical doers who can identify, conceptualise, formulate, design, develop, and produce modern systems drawing from a comprehensive tool-kit of multi-disciplinary engineering to solve problems.



Getting the Mechanics Right

You admire the designers of the plush cars that you come across when surfing the internet and yearn to join their trade? Mechanical Engineering is the field you should be considering to study in the university.

They are typically known as an integral part of teams involved with the development products and systems including automobiles, turbines, space shuttle vehicles, aircrafts, pumps, power plants, among others.

Their focus is often centered around the generation, distribution, and use of energy; the processing of materials; the control and automation of manufacturing systems; the design and development of machines; and the solutions to environmental problems.

Top Careers in Mechanical Engineering

Mechanical Engineers may be involved in a wide range of engineering jobs. However, some top job titles that may be directly related to the field include:

- ✦ Automotive Engineer
- ✦ Mechanical Engineer
- ✦ Biomedical Engineer
- ✦ Materials Engineer
- ✦ Robotics Engineer

Why ACity?

Home to
students from over
23 countries
across Africa

A fully digitized
campus with the fastest
internet speed
in a Ghanaian university

70%
practical lessons

**Experienced
Faculty**
with Industry Expertise

Every **ACitizen** has
the opportunity to do a
work placement
every vacation

Student-To-Faculty Ratio
5:1

Houses a
Digital Makerspace
and a **Technology &
Entrepreneurial Center**
for Creative freedom
& Innovation

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (English, Maths, Integrated/General Science) and 3 elective subjects. (Physics, Elective Maths + Chemistry OR any other elective relevant to the chosen Program)

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (English, Maths, Integrated/General Science) and 3 elective subjects. (Physics, Elective Maths + Chemistry OR any other elective relevant to the chosen Program)

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and 3 passes in the A-Levels. (Elective/Add/Further Maths and Physics mandatory)

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects. (Elective/Add/Further Maths and Physics mandatory)

FRENCH IB

Minimum of 50% overall average pass. (subject to approval NAB) (Maths, English and Physics mandatory)

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths, English and Physics mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills
French Language
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I
Introduction to Engineering
Introduction to Programming with Python
Physical Sciences
Pre-Calculus (with MATLAB)
Technology and Society

SEMESTER 2

Course Name

Analytic Geometry and Calculus I (with MATLAB)
Engineering Mechanics
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II
Introduction to Multidisciplinary Design
Logic and Critical Thinking
Programming in C
Sensors, Measurements and Instrumentation
Text and Meaning

SEMESTER 3

Course Name

Analytic Geometry and Calculus II (with MATLAB)
Computer-Aided Design and Manufacturing (CAD and CAM)
Electronics and Microprocessors
Engineering Thermodynamics
Fundamentals of Innovation and Entrepreneurship (FIE) I
Introduction to Material Science and Engineering
Leadership Seminar I

SEMESTER 4

Course Name

African Studies
Applied Linear Algebra (with MATLAB)
Fluid Mechanics
Fundamentals of Innovation and Entrepreneurship II
Manufacturing Processes I
Mechanics of Machines I
Probability, Statistics and Reliability (with MATLAB)

SEMESTER 5

Course Name

Differential Equations (with MATLAB)
Leadership Seminar II
Machine Design
Manufacturing Processes II
Mechanics of Machines II
Mechanics of Materials
Numerical Methods (with MATLAB)
Systems Dynamics

SEMESTER 6

Course Name

Applied Hydraulics and Pneumatics
Automatic Control System
Gas Dynamics and Jet Propulsion
Industry Internship
Internal Combustion Engines and Steam Turbine
Mechatronics
Project Management, Engineering Economics and Risk Analysis

SEMESTER 7

Course Name

Industrial Automation and Robotics

Project Phase I

Quality Control and Reliability
Engineering

Technical Elective - I

Technical Elective - II

Thermal Engineering

SEMESTER 8

Course Name

Environmental Science and
Engineering

Maintenance Engineering

Professional Ethics and Values

Project Phase II

Technical Elective III

Technical Elective - IV

ELECTIVES

Aerospace Engineering

Advanced Control Systems
Computational Fluid Mechanics
Composite Materials and Structures
Mechanical Vibrations

Biomedical Engineering

Biofluid Mechanics
Biomechanics
Biomaterials
Rehabilitation Engineering

Design and Manufacturing

Design of Jigs, Fixtures and Tools
Unconventional Machining Process
Composite Materials and Structures
Computer Integrated Manufacturing

Energy Systems

Heat and Mass Transfer
Refrigeration and Air Conditioning
Renewable Energy
Thermal Turbo Machines

Mechanical Design

Advanced CAD and CAM
Design of Machine Elements
Design of Jigs, Fixtures and Tools
Modelling and Analysis of Mechatronic
Systems

Thermo-Fluids Systems

Computational Fluid Dynamics
Refrigeration and Air Conditioning
Fluid Machinery
Heat and Mass Transfer

Robotics

Advanced CAD and CAM
Design of Transmission Systems
Advanced Control Systems
Modelling and Analysis of Mechatronic
Systems